Hazard Analysis
Software Engineering

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Table 1: Revision History

Date	Developer(s)	Change
Monday October 19, 2026	Harmanpreet, Anindita, Shivya, Vanessa	Initial Hazard Analysis Version
Date2	Name(s)	Description of changes
...	...	...

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1 Introduction

A hazard is a system condition or failure that could contribute to an undesirable outcome, resulting in losses like inaccurate accreditation assessments, poor scheduling suggestions, a loss of course information, or sensitive data breaches.

The CAS Course Deadline Management System is intended to support the Department of Computing and Software (CAS) at McMaster University in reviewing CAS course outlines for compliance with applicable university policies [1] and Canadian Engineering Accreditation Board (CEAB) requirements [2]. The system also analyzes assessment dates across courses and suggests alternative dates that reduce overlapping student workloads, while leaving final scheduling decisions to instructors. The system's requirements are specified in the project's Software Requirements Specification (SRS) [3].

This document identifies potential hazards associated with the system, examines their consequences, and establishes a basis for developing safety and security requirements to mitigate these risks. Section 2 defines the scope of the analysis and the losses it considers. Section 3 sets the system boundary and divides the system into components, and Section 4 lists the critical assumptions. Section 5 applies Failure Mode and Effect Analysis to each component, and Section 6 lists the resulting safety and security requirements. Section 7 states which of these requirements will be implemented during the capstone project.

2 Scope and Purpose of Hazard Analysis

The purpose of this hazard analysis is to identify potential failures in the CAS Course Deadline Management System that could negatively affect accreditation compliance reviews, course scheduling, or the handling of course information. The analysis covers the processing of course outlines, evaluation against McMaster policies and CEAB requirements, deadline conflict detection, the generation of scheduling suggestions, user access, and the storage and export of course information. The scope is limited to courses offered by the CAS department (for example, SFWRENG, COMPSCI, and MECHTRON courses) and excludes first-year courses, courses outside the department, and the security of the hosting infrastructure itself.
Potential losses resulting from system hazards include:

- A McMaster policy or CEAB violation in a course outline goes undetected and is not corrected.
- Increased student workload conflicts due to overlapping assignment and midterm deadlines, contributing to student stress, less time to prepare for assessments, and more McMaster Student Absence Form (MSAF) requests.
- An instructor moves an assessment to an unsuitable date, such as one that creates a new conflict or falls in a restricted period, because of an incorrect suggestion.
- Loss or corruption of course information, resulting in inaccurate course outline data and scheduling issues.
- Unauthorized disclosure of confidential information, including the master course sheet, compliance findings, and tentative assessment dates.
- Wasted administrator and instructor time due to false compliance findings, incorrectly identified deadline conflicts, or inaccurate recommendations.
- Loss of trust in the system due to repeated inaccurate or unreliable results, potentially leading users to stop using the system and reducing its intended benefits.

3 System Boundaries and Components

The CAS Course Deadline Management System processes CAS course outlines to support policy and accreditation compliance reviews and to suggest alternative assessment dates. Its boundary includes the components responsible for processing course information, evaluating policy and accreditation requirements, detecting scheduling conflicts, generating scheduling suggestions, and managing stored information. The system uses course outlines, a master course sheet for graduate attribute checks, and the academic calendar to identify restricted scheduling periods. The large language model (LLM) service, hosting infrastructure, users' web browsers, and instructors who prepare course outlines are outside the system boundary, but their interactions with the system and the information they provide are considered in the hazard analysis.

- **C1 User Interface:** Allows CAS administrators and instructors to log in, upload course outlines, review and confirm extracted information, enter tentative dates, and request and decide on scheduling suggestions.
- **C2 Course Information Database:** Stores course outline information, extracted assessment details, compliance findings, and tentative assessment dates.
- **C3 PDF Text Extraction:** Extracts text from uploaded PDF course outlines using a text-extraction library.
- **C4 LLM Information Structuring:** Uses an external LLM service to interpret extracted text and structure relevant information, including assessments, marking schemes, due dates, learning outcomes, and the graduate attributes each outcome addresses.
- **C5 Accreditation Compliance Analysis:** Evaluates course outline information against applicable McMaster policies and CEAB requirements, including graduate attribute checks using the master course sheet.
- **C6 Conflict Detection:** Analyzes assessment dates across courses to identify overlapping deadlines and potential student workload bottlenecks, taking restricted scheduling periods into account where applicable.
- **C7 Scheduling Suggestions:** Uses a CP-SAT solver [4] to generate proposed assessment dates based on scheduling constraints, identified conflicts, and applicable restricted periods.
- **C8 Results Presentation:** Presents compliance findings, detected conflicts, and proposed scheduling changes to administrators and instructors for review.
- **C9 Authentication and Access Control:** Authenticates users and enforces permissions to restrict access to system functions and confidential information.
- **C10 Data Export:** Exports permitted course information, compliance findings, and scheduling results for authorized users.
- **C11 External Dependencies:** Includes external services and software dependencies, specifically the LLM service used to structure extracted course outline information. The hosting infrastructure and users' web browsers are also external to the system boundary.

4 Critical Assumptions

The hazard analysis is based on the following assumptions about the system's operating environment, users, and input data. These assumptions establish the context for the FMEA. Where an assumption may not hold, the FMEA still includes the failure and its mitigation (for example, H5-4 and H6-5 cover an incorrect master course sheet and academic calendar).

- **Hosting Infrastructure and Recovery:** CAS IT is responsible for securing, backing up, and restoring the hosting infrastructure, including the application database and stored course information. The project team is responsible for documenting which application data must be included in those backups and verifying, with CAS IT, that a representative restore preserves confirmed assessment details, compliance findings, and tentative dates. The security of the hosting infrastructure itself remains outside the scope of this analysis.
- **User Accounts:** Users access the system using valid McMaster accounts and do not share their credentials.
- **Master Course Sheet:** The department-provided master course sheet contains correct information for the graduate attribute checks (SRS assumption A7 [3]).
- **Academic Calendar:** The academic calendar dates entered into the system are correct for the term being analyzed (SRS assumption A6).
- **Course Outline Confidentiality:** Course outlines are treated as public documents (SRS assumption A5). Other stored information, including the master course sheet, compliance findings, and tentative assessment dates, may be confidential.
- **Policy and Accreditation Requirements:** The McMaster policies and CEAB requirements maintained in the system reflect the versions in force for the term being reviewed. Updates are made when applicable requirements change (SRS likely changes LC3 and LC4).

5 Failure Mode and Effect Analysis

This section applies Failure Mode and Effect Analysis (FMEA) to each component identified in the System Boundaries and Components section (C1 to C11). For each component, the team asked how it could fail, what loss each failure would lead to, why it could happen, and how it would be noticed. Table 4 records the result. Each row gives:

- **Ref:** a unique identifier for the hazard, in the form H*component-row*, where *component* is the component number (H4-2 is the second hazard of C4), followed by the hazard's category and severity (Tables 2 and 3). The safety and security requirements refer back to these identifiers.
- **Failure Mode, Effects of Failure, Causes of Failure:** what goes wrong, the loss it leads to, and why it could happen.
- **Recommended Action:** how the failure is prevented or its effect is limited.
- **Req.:** the requirement that implements each recommended action. Where a row has more than one action, the actions and their requirements are lettered a., b., and so on. FR, NFR, DR, AIR and A refer to the functional, nonfunctional, data and AI requirements and the assumptions in the SRS [3]. SSR refers to the safety and security requirements in this document (Table 5). "New" marks an action that does not yet have a requirement; these are listed after the table.

Table 2: Hazard Categories

Code	Category
DQ	**Decision quality.** The system shows an incorrect or incomplete result (a date, a conflict, a finding, or a suggestion) that a user may act on.
DI	**Data integrity.** Stored data is lost, corrupted, or attached to the wrong course.
SE	**Security and privacy.** Data is seen or changed by someone who is not authorized to see or change it.
AV	**Availability.** A function cannot be used when it is needed.
HF	**Human factors.** The way people use or react to the system causes the loss, including over-trust, alert fatigue, exclusion, and stress.

Table 3: Hazard Severity

Severity	Definition
H (High)	The failure can lead directly to a loss listed in the Scope and Purpose section, such as an undetected policy or CEAB violation, an assessment moved to an unsuitable date, disclosure of confidential data, or loss of stored data, without being caught by user review.
M (Medium)	The failure leads to a loss only if a user also misses it, or it delays or degrades work in a way that can be recovered.
L (Low)	The failure causes inconvenience or wasted effort, but no incorrect result reaches a decision.

Ref	Failure Mode	Effects of Failure	Causes of Failure	Recommended Action	Req.
C1: User Interface					
H1-1 *HF, H*	Reviewer confirms extracted values without checking them	Incorrect dates or weights enter the calendar, conflict detection, and compliance results	a. Over-trust in automated output b. Long review lists	a. Show the source passage beside each value b. Require explicit confirmation, never by default; show flagged fields first	a. FR5, AIR4 b. FR5
H1-2 *DQ, H*	Unconfirmed extracted data is shown or used	Unchecked values appear in the calendar and analysis as if they were correct	Review status not checked before use	Use only confirmed values in the calendar, conflict detection, and compliance checking	FR5, AIR6
H1-3 *HF, M*	Interface cannot be used with a keyboard or screen reader	Staff with disabilities cannot use the system; AODA requirements [5, 6] are not met	Inaccessible interface components	Build and verify the interface to WCAG 2.0 Level AA [7]	NFR9
C2: Course Information Database					

Ref	Failure Mode	Effects of Failure	Causes of Failure	Recommended Action	Req.
H2-1 *DI, H*	Stored data is lost or corrupted	Confirmed data and decisions are lost and review must be repeated	a. Hardware or software failure b. Failed update	a. Have CAS IT back up the application database and stored files regularly; document the required data and verify a representative restore b. Keep records for the retention period	a. SSR9 b. DR4
H2-2 *DI, M*	Uploading an outline again replaces confirmed values	Reviewed values are replaced by unreviewed ones	New upload overwrites existing records	a. Preserve confirmed values until replacement is explicitly approved by the user b. Review newly extracted values before replacement	a. New: confirm before replace b. FR5
H2-3 *DI, M*	An outline is attached to the wrong course or term	Assessments are counted in the wrong cohort or term	a. Course code misread b. Wrong course selected	Show the course code and term on the review screen for confirmation	FR5
H2-4 *DQ, M*	A cross-listed course is stored more than once	Its assessments are counted twice, creating false clusters	Each course code treated as a separate course	Store one course with several course codes	A2
H2-5 *DI, L*	A change cannot be traced to who made it	Errors cannot be traced or corrected, and disputes cannot be resolved	Changes not recorded	Record the user, time, and previous value of each change	SSR6
C3: PDF Text Extraction					
H3-1 *AV, M*	Little or no text is extracted from an outline	Assessments for the course are missing from all results	Scanned or image-only PDF	a. Report the problem and keep the outline b. Allow the values to be entered manually	a. SSR1 b. NFR7
H3-2 *DQ, H*	Extracted text is out of order	A date or weight is attached to the wrong assessment	Tables and multi-column layouts	a. Show the source passage for each value b. Test on outlines with a range of layouts	a. FR5, AIR4 b. NFR1
H3-3 *SE, M*	An unsupported, oversized, or malicious file is uploaded	Processing fails or slows the system; a malformed file exploits the PDF library	No file checks before processing	Accept only PDF files smaller than 5 MB; reject others before processing	SSR2
C4: LLM Information Structuring					

Ref	Failure Mode	Effects of Failure	Causes of Failure	Recommended Action	Req.
H4-1 *DQ, H*	The model reports a value that is not in the outline	A false date or weight is shown and may be confirmed	Model fills gaps or rewrites text	a. Reject values without a source passage b. Require review	a. AIR4, AIR5 b. FR5
H4-2 *DQ, H*	The model omits an assessment or a field	Conflicts are missed and weights do not total	Unusual wording or layout	a. Flag missing fields b. Flag weights that do not total c. Measure on the accuracy test set	a. FR4 b. FR13 c. NFR1
H4-3 *DQ, M*	A date marked TBD, or too vague to resolve, is turned into a date	A guessed date appears in the calendar as if it were real	Model resolves ambiguity instead of reporting it	Report such dates as missing and never guess	AIR5, FR4
H4-4 *DQ, M*	The model links a learning outcome to the wrong graduate attribute	A graduate attribute is reported as covered or missing when it is not	a. Attribute not listed by number or name b. Ambiguous learning outcome wording	a. Give the model the CEAB graduate attribute definitions b. Record whether each attribute was listed in the outline or identified by the model c. Require review of identified attributes before the compliance check uses them	a. AIR1 b. AIR3 c. FR5, AIR6
H4-5 *SE, M*	Text in an outline changes the model's behaviour (prompt injection)	Manipulated values or findings are shown	Outline text treated as instructions	a. Treat outline text as data only b. Check the output against the expected structure c. Keep human review	a. New: output validation b. New: output validation c. FR5
H4-6 *SE, H*	Confidential data is sent to an external model service	Confidential department data, such as the master course sheet, leaves McMaster's control	Confidential data included in a request to the model by mistake	a. Send only public outline text and the CEAB attribute definitions to an external service b. Compare graduate attributes with the master course sheet in the system's own code	a. AIR1, AIR2, DR2 b. FR14, DR1
C5: Accreditation Compliance Analysis					
H5-1 *DQ, H*	A real policy or CEAB violation is missed	The violation is not corrected	a. Rule not encoded b. Extraction error	a. List the rules checked for each outline b. Require review of extracted values c. Measure on a test set of outlines with known issues	a. SSR5 b. FR5 c. FR13, FR14, NFR1

Ref	Failure Mode	Effects of Failure	Causes of Failure	Recommended Action	Req.
H5-2 *HF, L*	A violation is reported that does not exist	Staff waste time, and trust in the system falls	a. Extraction error b. Rule too strict	Show the rule and the source text for each finding	FR13, AIR4
H5-3 *DQ, M*	A result with no findings is read as full compliance	Issues outside the encoded rules are assumed not to exist	Only some rules can be checked automatically	State which rules were and were not checked	SSR5
H5-4 *DQ, M*	The master course sheet is wrong or out of date	Graduate attribute findings are wrong	Department changes the expected attributes (LC4)	a. Show which version of the sheet each check used b. Only administrators change it	a. New: reference version shown b. A7, SSR7
H5-5 *DQ, M*	A course is missing from the master course sheet	No graduate attribute findings are reported, which reads as compliant	Electives and some programs are not on the sheet	State that the course was not checked	SSR5
H5-6 *DQ, M*	Policy rules are out of date	Outlines are checked against an old version of the policy	Policy revised (LC3)	a. Record the policy version used for each finding b. Review the rules when the policy changes	a. New: reference version shown b. LC3
C6: Conflict Detection					
H6-1 *DQ, M*	A cohort is derived incorrectly from course codes	Clusters are missed, or reported for students who are not affected	a. Unusual course codes b. Electives grouped with core courses (A3)	a. Test the derivation on all CAS course codes b. Report codes that cannot be read	a. A1, FR8 b. A1
H6-2 *DQ, M*	A missing or TBD date hides a conflict	A real cluster is not shown	Dates not yet set	a. Show the number of undated assessments with each result b. Prompt the instructor for a tentative date	a. FR4 b. FR6
H6-3 *DQ, M*	Analysis is not re-run after a date changes	Out-of-date conflicts are shown	Re-analysis not triggered	Re-run analysis on every change and show when it last ran	FR10
H6-4 *DQ, H*	An assessment in a restricted period is not detected	A policy violation goes unnoticed	Restricted period missing or wrongly encoded	a. Check every assessment against each restricted period b. Test each type of restricted period	a. FR9 b. NFR1

Ref	Failure Mode	Effects of Failure	Causes of Failure	Recommended Action	Req.
H6-5 *DQ, H*	The academic calendar is wrong or out of date	Restricted-period checks and suggestions are wrong for every course	a. Entry error b. Previous term's dates reused	a. Show which calendar each analysis used b. Confirm the calendar each term; only administrators change it	a. New: reference version shown b. A6, SSR7
C7: Scheduling Suggestions					
H7-1 *DQ, H*	A suggested date falls on a holiday or in a restricted period	An instructor moves an assessment to a date that breaks policy	Restricted periods not passed to the solver	Apply the applicable policy restrictions when determining eligible assessment dates, considering assessment type, weight, and restricted periods	SSR3
H7-2 *DQ, M*	A suggestion is based on data that has since changed	The accepted date creates a new conflict	Other dates change before the instructor decides	Mark the suggestion as out of date and offer to recompute	SSR4
H7-3 *DQ, M*	A suggested date creates a conflict for another cohort	The workload moves to other students instead of being reduced	Cross-listed course belongs to several cohorts (A2)	Check every cohort each course code belongs to before suggesting a date	A2, FR11
H7-4 *AV, L*	No feasible date is found, or the solver times out	The instructor receives no suggestion	Constraints cannot all be met	a. State that no suitable date was found and why b. Never return a date that breaks a constraint	a. FR11 b. SSR3
H7-5 *DQ, M*	The explanation does not match the data	The instructor is misled about why a date is suggested	Explanation generated without the solver's figures	Build the explanation only from the solver's figures, and show those figures beside it	FR11
H7-6 *DQ, H*	An assessment date is changed without the instructor's approval	The instructor's authority over the course is overridden	Suggestion applied automatically	Never change a date without explicit approval	FR12
C8: Results Presentation					
H8-1 *DQ, M*	A tentative date is shown as if it were confirmed	Instructors plan around a date that may change and judge conflicts wrongly	Display does not distinguish date status	Mark tentative dates in the calendar and in exports	FR7
H8-2 *HF, M*	Too many workload warnings are shown	Users ignore warnings and real conflicts are missed; instructors feel pressured	a. Threshold set too low b. All same-level courses grouped as one cohort (A3)	a. Rank clusters by size and show the largest first b. Tune *CLUSTER_THRESHOLD* with the CAS supervisors	a. New: warning ranking b. FR8

Ref	Failure Mode	Effects of Failure	Causes of Failure	Recommended Action	Req.
C9: Authentication and Access Control					
H9-1 *SE, H*	A user reaches data without logging in	Confidential data, such as the master course sheet, findings, and tentative dates, is exposed	An endpoint without an authentication check	Require login for every request to data	FR15, NFR8
H9-2 *SE, H*	An instructor changes another instructor's course	Assessment dates are changed without the authority to do so	Missing or incorrect role check	a. Enforce role checks on the server for every change b. Record each change	a. FR16, DR3 b. SSR6
H9-3 *SE, M*	A session is left open on a shared computer	Another person views or changes data as that user	No inactivity timeout	End a session after a period of inactivity	New: session timeout
H9-4 *SE, H*	A non-administrator changes the master course sheet or academic calendar	Compliance and restricted-period results become wrong for every course	Reference data editable by all roles	a. Allow only administrators to change reference data b. Record each change	a. SSR7 b. SSR6
C10: Data Export					
H10-1 *SE, M*	An export contains confidential or unconfirmed data	Confidential data leaves the system, or unchecked values are used elsewhere	Export not limited by role or review status	a. Export only confirmed data that the user's role may see b. Mark tentative dates	a. FR17, DR3; New: export limits b. FR7
H10-2 *SE, L*	A cell in the exported file runs as a formula when opened	A harmful formula runs in the user's spreadsheet program	Outline text that begins with =, +, −, or @	Escape cells that begin with these characters	New: CSV escaping
H10-3 *DI, L*	Characters in the export are garbled	Course names or text cannot be read	Wrong character encoding	Export in UTF-8 and test in common spreadsheet programs	FR17
C11: External Dependencies					
H11-1 *AV, M*	The language model service is unavailable	Extraction and suggestion explanations are delayed	a. Service outage b. Network failure	a. Keep the outline, inform the user, and allow a retry b. Keep other functions available	a. SSR1 b. NFR7

H11-2 *DQ, M*	A change of model or model version lowers accuracy	More extraction errors reach reviewers without anyone noticing	a. Provider updates the model b. Team switches model (LC1)	Re-run the accuracy test set before any model change is deployed	NFR1; New: model change test

Table 4: Failure Mode and Effect Analysis

The analysis identified the following recommended actions that do not yet have a requirement. Each will be added to the Safety and Security Requirements (Table 5) and to the SRS:

- **Warning ranking** (H8-2): rank workload clusters by size.
- **Session timeout** (H9-3): end a session after inactivity.
- **Model change test** (H11-2): re-run the accuracy test set before a model change is deployed.
- **Output validation** (H4-5): check model output against the expected structure.
- **Confirm before replace** (H2-2): keep confirmed values when an outline is uploaded again.
- **Reference version shown** (H5-4, H5-6, H6-5): record which calendar, master course sheet, and policy version each result used.
- **Export limits** (H10-1): export only confirmed data the user's role may see.
- **CSV escaping** (H10-2): escape exported cells that could run as formulas.

Row H4-6 also led to a change in the SRS: AIR1 now excludes the master course sheet from the model's input, and the graduate attributes are compared with the sheet by the system's own code.

6 Safety and Security Requirements

The requirements below are in addition to safeguards already in the SRS [3] (see Table 4 of the SRS), including user review of extracted data (FR5), flagging of missing fields (FR4), instructor approval of date changes (FR12), login (FR15), role-based access (FR16), and restrictions on data sent to external services (AIR2). Additional safety and security requirements are listed in Table 5. Public outline text may be sent to our LLM, but confidential data and student information shall remain internal (see DR2).

Table 5: Safety and Security Requirements

ID	Hazard(s)	Requirement
SSR1	H3-1, H11-1: outline processing failure resulting in an invalid or incomplete schedule or lost data	The system shall display an actionable error when a course outline PDF cannot be read or the model fails to process it, shall not create assessments for that outline automatically, and shall keep all data already entered.
SSR2	H3-3: unsupported or malicious input file	The system shall accept only PDF files smaller than 5 MB, and shall reject any other file before processing it.
SSR3	H7-1, H7-4: assessment scheduled during a restricted period	The system shall use the academic calendar and McMaster scheduling policies to prevent invalid date recommendations. During the test and examination ban period, tests and examinations worth more than 10% of the course grade shall not be suggested. Assessments shall also not be scheduled during recesses or other policy-restricted periods.
SSR4	H7-2: out-of-date scheduling suggestion	The system shall mark a date suggestion as out of date if the assessment's date, or the other assessments in the affected weeks, change before the instructor decides on it.
SSR5	H5-1, H5-3, H5-5: incomplete compliance check or falsely reported absence of findings	The system shall show which compliance rules were checked for each outline, and shall state that a course was not checked when it is missing from the master course sheet, rather than reporting no findings.
SSR6	H2-5, H9-2, H9-4: unauthorized or untraceable data modification	The system shall record the user, the time, and the previous value whenever an assessment date, an extracted value, or reference data (the master course sheet or the academic calendar) is changed, and shall retain that record for the retention period in DR4.
SSR7	H5-4, H6-5, H9-4: unauthorized modification of course-rule data	The system shall allow only administrators to change the master course sheet and the academic calendar.
SSR8	H2-2: overwriting previously confirmed course information	The system shall preserve previously confirmed course information when an outline is re-uploaded until newly extracted values have been reviewed and explicitly approved by the user for replacement.
SSR9	H2-1: loss or corruption of stored course information	CAS IT shall maintain regular backups of the application database and stored files. Before production use and after material schema changes, the project team and CAS IT shall verify a representative restore containing confirmed assessment details, compliance findings, and tentative dates. The restore procedure and verification results shall be documented.

7 Roadmap

The capstone implementation will prioritize safety and security requirements that prevent data loss, restrict unauthorized changes, and reduce the risk of incorrect assessment scheduling. Requirements that cannot be completed within the capstone timeline will be documented as future work.

SSR9 is an external operational dependency on CAS IT. The project team will document the application's backup contents and restore procedure and will coordinate a representative restore verification with CAS IT before production use. CAS IT will remain responsible for the backup schedule, backup storage, infrastructure restoration, and ongoing recovery operations. The project team will not manage the hosting infrastructure or operate its production backup service.

During the capstone timeline, the team will implement SSR1 to handle PDF extraction and model processing failures without losing previously entered data or automatically creating assessments from an unsuccessfully processed outline. SSR2 will restrict uploads to PDF files smaller than 5 MB and reject unsupported files before processing. The team will also implement SSR3 to prevent assessment date suggestions that conflict with holidays, recess, examination periods, or test and examination ban periods defined by the academic calendar. SSR4 will ensure that suggestions are marked as out of date when relevant assessment dates change before instructor review.

The team will also prioritize SSR5 to identify which compliance rules were checked for each course outline and flag courses that are missing from the master course sheet. SSR7 will restrict modifications to the master course sheet and academic calendar to administrators, complementing the existing login and role-based access controls.

If time permits, the team will implement SSR6 to record the user, timestamp, and previous value whenever assessment dates, extracted values, or reference data, including the master course sheet and academic calendar, are changed. Audit records will be retained for the period specified in DR4. If this requirement cannot be completed within the capstone timeline, it will be documented as future work.

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8 Appendix — Generative AI Use Disclosure

8.1 AI Tools Used

The following AI tools were used:

- Grok (version 4.7)
- Claude (Anthropic), Claude Opus 5.5 model
- ChatGPT (OpenAI), GPT-4-turbo model
- GitHub Copilot (version 1.79.0)

8.2 How and Where Used

The tools were used in the following manner:

- Grok was used to fix formatting issues and address pull request comments in this Hazard Analysis document, in Safety and Security Requirements section.
- Claude was was used to review and refine the document, suggesting improvements to structure and clarity, and assisting with grammar and style. It was used to primarily refine the FMEA table and the Safety and Security Requirements section.
- ChatGPT and Copilot were used for formatting and grammar suggestions, and to assist with the writing of the Roadmap section.

8.3 Verification of AI-Generated Content

Any output by LLM models was reviewed thoroughly and modified before its suggestions were added to the document. Course materials and deliverable requirements were manually reviewed to ensure that all technical assertions, requirements, and references were accurate and complete.

9 Appendix — Reflection

1. What went well while writing this deliverable?

When writing this deliverable, it was relieving for our team to consolidate and organize all the potential risks we thought of on paper. We were able to brainstorm various hazards, as well as critical assumptions thanks to the discussions we've had with stakeholders prior to the deliverable.

2. What pain points did you experience during this deliverable, and how did you resolve them?

A major pain point our team experienced while working on the Hazard Analysis document was considering all the potential risks that could occur before even having a working prototype of our project. With hardware, it is easy to imagine the potential hazards involved with the final product, but software requires a good visualization of what risks can occur from the perspective of the user. Depending on the technological skill level, different users may experience vastly different situations when interacting with our system. However, to resolve this we broke down our system into various components based on the user process (what might they do in chronological order). This helped us to understand where hidden hazards could be located, as well as any assumptions we may have made due to our own expertise with technology.

3. Which of your listed risks had your team thought of before this deliverable, and which did you think of while doing this deliverable? For the latter ones (ones you thought of while doing the Hazard Analysis), how did they come about?

Before the Hazard Analysis deliverable, some potential risk ideas intuitively came to our group while we were brainstorming how to go about developing a prototype of our design, such as potentially leaking sensitive and confidential information (ex. student information). On the other hand, other risks that were only considered while working on the deliverable included submitting malicious file inputs and untraceable data modifications, which we thought of after considering how a user would interact with the system step-by-step. Additionally, discussing with our stakeholders helped us gain different perspectives and better understand what users might be concerned about.

4. Other than the risk of physical harm (some projects may not have any appreciable risks of this form), list at least 2 other types of risk in software products. Why are they important to consider?

One type of risk in software products is cybersecurity risks and potential data leaks. In our project, while we aren't working with critically sensitive information, we need to consider any administrative or personal information, as well as ensuring that authorized users are able to interact with the system without giving access to non-authorized users. This is important because data breaches can cause harm by exposing personal information to people who may have malicious intentions without consent.

Another type of software risk is misinformation and/or component failures when a user is interacting with the system. Not only is this frustrating for the user, but this can result in the failure of the complete product. One of the first things our team recognized with our project was that the last thing our users would want is to navigate a software platform that has a high learning curve or false information, as they are not expected to have as high technical knowledge as we do. Therefore, designing a system that is both simple and provides accurate data is essential to the success of our project.